Final Year Project Confirmation Letter

To

The Head of the Department
Department of Computer Science
Godavari Global University

Date: _____

Subject: Approval of Final Year Project Title and Abstract – Review 0

Respected Sir/Madam,

We, the undersigned final-year students, are submitting our proposed project title and abstract for our Final Year Project (Review 0).

Proposed Project Title: AHD-Jailbreak: Adaptive Hypergraph Defense for Detecting Evolving Jailbreak Attacks in Large Language Models

Abstract:

Large Language Models (LLMs) are increasingly deployed in applications that require reliable and safe responses, yet they remain vulnerable to evolving jailbreak attacks designed to bypass safety constraints through prompt manipulation, multi-turn interactions, and novel attack strategies. This project proposes AHD-Jailbreak, an adaptive hypergraph-based defense framework that models complex relationships among user prompts, conversational context, semantic features, safety constraints, and emerging attack patterns. Unlike static defense mechanisms that primarily depend on previously known attack structures, the proposed adaptive approach is designed to identify anomalous or previously unseen relationships and update its risk representation over time, enabling improved resilience against evolving jailbreak strategies.

The proposed framework is intended to operate as a lightweight wrapper layer that can be placed in front of different AI or LLM systems, making the approach model-agnostic rather than dependent on a single provider or model architecture. The defense layer aims to identify potentially policy-violating or unsafe requests before they are forwarded for LLM inference, allowing some unsafe interactions to be blocked or handled locally without unnecessary model routing and associated inference cost. The research will evaluate the adaptive hypergraph against static hypergraph-based and conventional defense baselines using known and held-out attack categories, measuring robustness to unseen attacks, detection effectiveness, false-positive rate, computational overhead, and the impact on legitimate user interactions.

We have attached the detailed abstract of our proposed project along with this letter for your reference.

We are ready to present our project concept through slides for Review 0. A working prototype of the system can also be demonstrated during the review.

Thanking you,
Yours sincerely,
1. Sesetti Pavan Kumar Swamy
2. Rajana Rajeswari

(Student batch details and Guide’s approval are provided on the next page)

Student Batch Details

S.No	Student Name	PIN / Register Number	Branch / Section
1	Sesetti Pavan Kumar Swamy	24555A0519	CSE – C
2	Rajana Rajeswari	23551A05C9	CSE – B

Guide’s Approval

I have reviewed the proposed title and abstract. I approve this project for Review 0.

Name of the Faculty Guide: Mrs. S. Suguna Sri

Designation: _____

Signature of the Guide: _____ Date: ____ / ____ / _____